UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	Path animation moves an object along a predefined route.	Path animation follows a motion path.
A2	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A3	FALSE	3D animation does not use modelling.	3D animation relies on models.
A4	TRUE	MPEG is a common family of video compression formats.	MPEG defines video compression standards.
A5	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.
A6	TRUE	A URL identifies the address of a web resource.	A URL is a resource locator.
A7	TRUE	Chroma key compositing replaces a colour background with another image.	Chroma key removes a selected colour.
A8	TRUE	HTML is used to structure content on web pages.	HTML marks up web content.
A9	FALSE	Interactivity level is irrelevant when choosing an authoring tool.	The required interactivity affects tool choice.
A10	FALSE	Streaming requires the whole file to download before playback.	Streaming can play while downloading.
A11	FALSE	Higher resolution always means a smaller file.	Higher resolution usually increases file size.
A12	FALSE	A vector image becomes pixelated when enlarged.	Vector images scale without pixelation.
A13	TRUE	Frame rate is the number of video frames displayed per second.	Frame rate is measured in fps.
A14	TRUE	Keyframes define important positions in an animation.	Keyframes mark major changes.
A15	TRUE	Sampling converts a continuous sound wave into digital values.	Sampling measures the wave at intervals.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	J	Bitmap	image made of pixels
B2	N	Frame rate	frames displayed per second
B3	O	Storyboard	visual sequence plan
B4	A	Vector	image made of mathematical objects
B5	E	Serif	font with small strokes at character ends
B6	G	Keyframe	important frame in an animation
B7	I	Tweening	generating in-between frames
B8	H	Scanner	input device that digitises documents
B9	M	RGB	additive colour model used by screens
B10	B	Cel animation	hand-drawn frame-by-frame animation
B11	K	Sampling	converting sound into digital values
B12	L	Plotter	output device for large drawings
B13	F	Multimedia	integration of text, graphics, audio, video and animation
B14	D	Target audience	intended users of a product
B15	C	Graphics tablet	input device for drawing

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	D	Non-linear
C2	D	Multimedia with hyperlinks
C3	B	A cinema presentation
C4	C	The user can control content and flow
C5	B	PNG
C6	C	Temporary working memory
C7	D	Assemble multimedia elements
C8	D	It combines visuals and sound
C9	D	Analog sound to digital values
C10	C	Text, graphics, audio, video and animation
C11	B	Merging of media and computing
C12	A	One shape into another
C13	D	Another resource
C14	C	Compiler
C15	D	Original creative works
C16	C	Drawing input
C17	B	Plotter
C18	C	Card-based
C19	C	Structure web content
C20	C	Smooth jagged edges
C21	B	Vector
C22	C	Musical notes and instructions
C23	D	Improves accuracy
C24	C	Adds digital information to the real world
C25	B	A visual sequence plan
C26	A	Display web pages
C27	B	Small strokes at character ends
C28	B	Frequency
C29	B	Links to other content
C30	B	Messages with a beginning and end

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PART D - WORKED ANSWERS

QUESTION 1

Topic: text

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

Sans serif is cleaner at screen sizes; serif aids reading in print.

(3 marks)

b) Describe TWO roles of hypertext in multimedia.

It links related content and supports non-linear navigation.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

Clear headings and links guide users through HTML pages and make the information easier to scan and follow.

(4 marks)

QUESTION 2

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(4 marks)

QUESTION 3

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(4 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

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QUESTION 4

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(5 marks)

END OF MARKING SCHEME

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